1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

Primary hypothesis: In tumors with activating PIK3CA alterations, PI3K/AKT/mTOR signaling upregulates PD-L1 (CD274) expression, creating a functional immune-suppressive tumor microenvironment that is not overcome by high TIL density alone. This renders high-TIL tumors with PI3K activation (and elevated PD-L1) associated with shorter progression-free interval (PFI) compared with high-TIL, PI3K-wild-type tumors. Secondary modifiers—FOLH1 (PSMA), TMPRSS2, and ERG—reflect stem-like/lineage programs that synergize with PI3K signaling to further enhance immune evasion and modulate PD-L1 regulation. Consequently, the prognostic value of TIL density is context-dependent and most adverse in PI3K-activated tumors with PD-L1 upregulation, with even greater effect in tumors harboring TMPRSS2/ERG fusion or elevated FOLH1 expression. Imaging-derived radiomic signatures of the immune microenvironment will correlate with this axis, serving as non-invasive proxies for PI3K-driven PD-L1–mediated suppression. Testable predictions include: (1) among TIL-high tumors, PIK3CA-altered cases exhibit higher CD274 expression and shorter PFI vs. PIK3CA-wild-type; (2) inhibition of PI3K signaling reduces PD-L1 expression and restores T cell–mediated cytotoxicity in vitro; (3) TMPRSS2/ERG and/or FOLH1 status potentiates PD-L1 upregulation and correlates with worse PFI in the TIL-high, PI3K-activated subset; (4) radiomic/imaging features indicative of an immunosuppressive microenvironment will be enriched in the PI3K-activated, TIL-high group and respond to PI3K/PD-1 axis modulation in preclinical models.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	PIK3CA, CD274
Primary endpoint	PFI

2. Why This Gene — Not Others

Primary hypothesis gene(s)

PIK3CA: PIK3CA alterations and PD-L1 as CD274.

CD274: CD274.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	FOLH1	Supporting modifier from knowledge graph

Secondary	TMPRSS2	Supporting modifier from knowledge graph
Secondary	ERG	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: In functional experiments, restoring or enhancing the anchor gene's signaling will improve immune-mediated tumor control in imaging-feature-positive models.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- FOLH1: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- TMPRSS2: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.